

Terraform Assignment

EC2 and S3 Resource Lifecycle

Provisioning and cleanup of AWS resources using Terraform

This document demonstrates provisioning a **t2.micro EC2 instance** and an **S3 bucket** in the **us-east-1** region using Terraform, validating both in the AWS Console, and destroying all resources with `terraform destroy`. Both resources are tagged with `Name = "TerraformAssignment"`.

1. terraform apply — Execution

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_key_pair.public-key: Creating...
aws_s3_bucket.assignment: Creating...
aws_security_group.allow-http-inbound: Creating...
aws_key_pair.public-key: Creation complete after 1s [id=id_rsa]
aws_security_group.allow-http-inbound: Creation complete after 5s [id=sg-02a5d653cffe528e5]
aws_instance.t2-micro: Creating...
aws_s3_bucket.assignment: Creation complete after 7s [id=terraform-assignment-202608220821097605000000001]
aws_instance.t2-micro: Still creating... [00m10s elapsed]
aws_instance.t2-micro: Creation complete after 16s [id=i-0d2dd3fc7667863f8]

Apply complete! Resources: 4 added, 0 changed, 0 destroyed.
(base) PS E:\Terraform S3 Bucket> |
```

Terminal output of terraform apply. Apply complete! Resources: 4 added, 0 changed, 0 destroyed.

2. EC2 Instance in AWS Console

The screenshot displays the AWS Management Console for an EC2 instance. The left sidebar shows the navigation menu with categories like EC2, Images, Elastic Block Store, Network & Security, Load Balancing, and Auto Scaling. The main content area shows the 'Instance summary for i-0d2dd3fc7667863f8 (TerraformAssignment)'. The instance is in the 'Running' state, type 't2.micro', and is located in 'us-east-1'. The 'Security' tab is selected, showing a table of inbound rules with columns for Name, Security group rule ID, Port range, Protocol, Source, Security groups, and Description. The table contains one rule named 'allow-http-inbound' with port range 80 and protocol TCP. The console also shows various other details like IAM role, Subnet ID, VPC ID, and Launch time.

Instance summary for i-0d2dd3fc7667863f8 (TerraformAssignment) info

Updated less than a minute ago

Instance ID: i-0d2dd3fc7667863f8

IPv6 address: -

Hostname type: IP name: ip-192-168-1-46.ec2.internal

Answer private resource DNS name: -

Auto-assigned IP address: 34.229.91.166 (Public IP)

IAM role: -

IMDSv2: Required

Operator: -

Public IPv4 address: 34.229.91.166 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-192-168-1-46.ec2.internal

Instance type: t2.micro

VPC ID: vpc-016d35616394c4728 (Gerald-VPC)

Subnet ID: subnet-049a177f90e7cfd1

Instance ARN: arn:aws:ec2:us-east-1:448513989308:instance/i-0d2dd3fc7667863f8

Private IPv4 addresses: 192.168.1.46

Public DNS: -

Elastic IP addresses: -

AWS Compute Optimizer finding: User: arn:aws:iam::448513989308:user/razaulagib2019@gmail.com is not authorized to perform: compute-optimizer:GetEnrollmentStatus on resource: 'because no identity-based policy allows the compute-optimizer:GetEnrollmentStatus action

Auto Scaling Group name: -

Managed: false

Details | Status and alarms | Monitoring | **Security** | Networking | Storage | Tags

▼ Security details

IAM role: -

Security groups: sg-02a5d53cfe532e5 (allow-http-inbound)

▼ Inbound rules

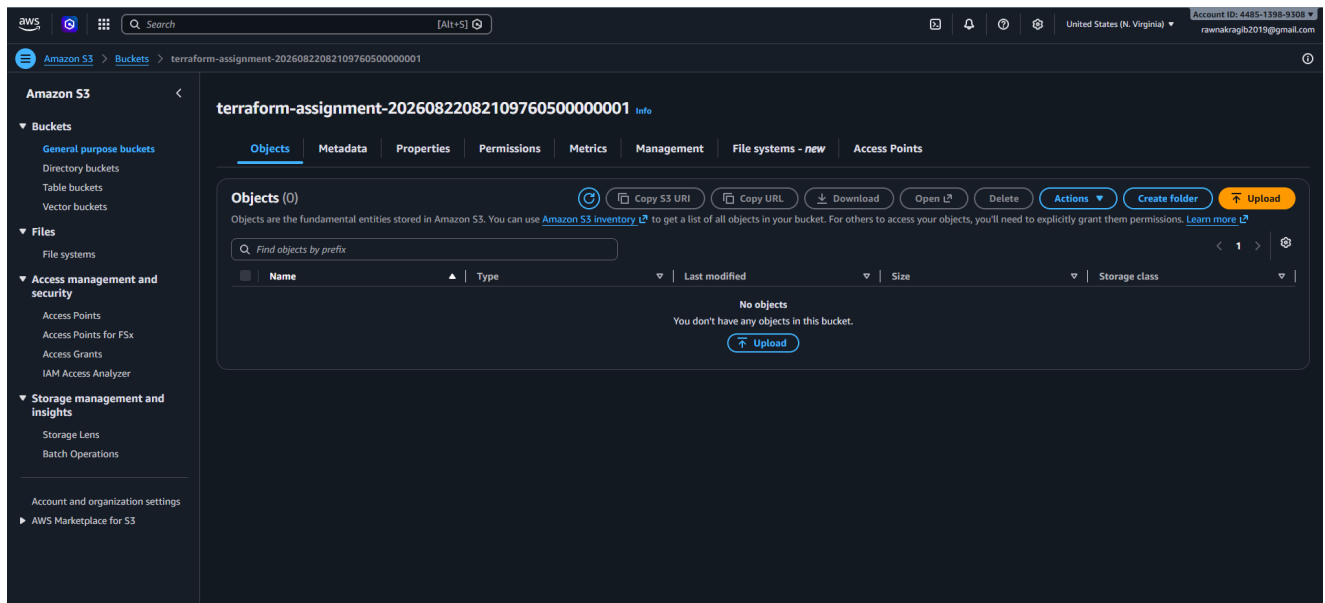
Q. Filter rules

Name	Security group rule ID	Port range	Protocol	Source	Security groups	Description
-	sg-02a5d53cfe532e5	80	TCP	0.0.0.0/0	allow-http-inbound	-

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Instance i-0d2dd3fc7667863f8 (TerraformAssignment) — type **t2.micro**, state **Running**, in region N. Virginia (us-east-1).

3. S3 Bucket in AWS Console



Bucket terraform-assignment-20260822082109760500000001 created with a unique auto-generated name.

4. terraform destroy — Successful Cleanup

```
Plan: 0 to add, 0 to change, 10 to destroy.

Do you really want to destroy all resources?
  Terraform will destroy all your managed infrastructure, as shown above.
  There is no undo. Only 'yes' will be accepted to confirm.

Enter a value: yes

aws_route_table_association.public-rt-assoc: Destroying... [id=rtbassoc-023574db121206e8c]
aws_route.default-route: Destroying... [id=r-rtb-0cfd78607e28dd3201080289494]
aws_s3_bucket.assignment: Destroying... [id=terraform-assignment-20260822082109760500000001]
aws_instance.t2-micro: Destroying... [id=i-0d2dd3fc7667863f8]
aws_s3_bucket.assignment: Destruction complete after 1s
aws_route_table_association.public-rt-assoc: Destruction complete after 1s
aws_route.default-route: Destruction complete after 2s
aws_internet_gateway.internet-access: Destroying... [id=igw-085c3ee8b59201ebc]
aws_route_table.public-rt: Destroying... [id=rtb-0cfd78607e28dd320]
aws_route_table.public-rt: Destruction complete after 2s
aws_instance.t2-micro: Still destroying... [id=i-0d2dd3fc7667863f8, 00m10s elapsed]
aws_internet_gateway.internet-access: Still destroying... [id=igw-085c3ee8b59201ebc, 00m10s elapsed]
aws_instance.t2-micro: Still destroying... [id=i-0d2dd3fc7667863f8, 00m20s elapsed]
aws_internet_gateway.internet-access: Still destroying... [id=igw-085c3ee8b59201ebc, 00m20s elapsed]
aws_internet_gateway.internet-access: Destruction complete after 21s
aws_instance.t2-micro: Destruction complete after 23s
aws_key_pair.public-key: Destroying... [id=id_rsa]
aws_subnet.public: Destroying... [id=subnet-049a177f50ed7cfd]
aws_security_group.allow-http-inbound: Destroying... [id=sg-02a5d653cffe528e5]
aws_key_pair.public-key: Destruction complete after 1s
aws_subnet.public: Destruction complete after 2s
aws_security_group.allow-http-inbound: Destruction complete after 2s
aws_vpc.Geralt: Destroying... [id=vpc-016d35616394c4728]
aws_vpc.Geralt: Destruction complete after 2s

Destroy complete! Resources: 10 destroyed.
(base) PS E:\Terraform S3 Bucket> |
```

Terminal output of terraform destroy. Destroy complete! Resources: 10 destroyed.

5. Terraform Configuration (main.tf)

```
terraform {
  required_providers {
    aws = {
      source = "hashicorp/aws"
      version = "~> 5.7.0"
    }
  }
}

provider "aws" {
  profile = "Geralt"
  region = "us-east-1"
}

data "aws_ami" "amazon_linux" {
  most_recent = true
  owners      = ["amazon"]

  filter {
    name   = "name"
    values = ["al2023-ami-*-x86_64"]
  }

  filter {
    name   = "virtualization-type"
    values = ["hvm"]
  }
}

resource "aws_vpc" "Geralt" {
  cidr_block = "192.168.0.0/16"
  tags = {
    Name = "Geralt-VPC"
  }
}

resource "aws_subnet" "public" {
  vpc_id = aws_vpc.Geralt.id
  cidr_block = "192.168.1.0/24"
  availability_zone = "us-east-1a"
}

resource "aws_route_table" "public-rt" {
  vpc_id = aws_vpc.Geralt.id
  tags = {
    Name = "public-route-table"
  }
}

resource "aws_internet_gateway" "internet-access" {
  vpc_id = aws_vpc.Geralt.id
  tags = {
    Name = "main-igw"
  }
}

resource "aws_route" "default-route" {
  route_table_id = aws_route_table.public-rt.id
  destination_cidr_block = "0.0.0.0/0"
  gateway_id = aws_internet_gateway.internet-access.id
}

resource "aws_route_table_association" "public-rt-assoc" {
  subnet_id = aws_subnet.public.id
  route_table_id = aws_route_table.public-rt.id
}

resource "aws_security_group" "allow-http-inbound" {
  name = "allow-http-inbound"
  description = "sg for allowing inbound traffic"
  vpc_id = aws_vpc.Geralt.id
}
```

```

    ingress {
      from_port = 80
      to_port = 80
      protocol = "tcp"
      cidr_blocks = ["0.0.0.0/0"]
    }
    egress {
      from_port = 0
      to_port = 0
      protocol = "-1"
      cidr_blocks = ["0.0.0.0/0"]
    }
  }

  resource "aws_key_pair" "public-key" {
    key_name = "id_rsa"
    public_key = file("C:/Users/User/.ssh/id_rsa.pub")
  }

  resource "aws_instance" "t2-micro" {
    ami = data.aws_ami.amazon_linux.id
    instance_type = "t2.micro"
    key_name = aws_key_pair.public-key.key_name
    subnet_id = aws_subnet.public.id
    vpc_security_group_ids = [ aws_security_group.allow-http-inbound.id ]
    associate_public_ip_address = true
    tags = {
      Name = "TerraformAssignment"
    }
  }

  resource "aws_s3_bucket" "assignment" {
    bucket_prefix = "terraform-assignment-"
    tags = {
      Name = "TerraformAssignment"
    }
  }
}

```